BigEarthNet.txt: A Large-Scale Multi-Sensor Image-Text Dataset and Benchmark for Earth Observation

Johann-Ludwig Herzog[*1,2], Mathis Jürgen Adler[*1,2], Leonard Hackel[1,2], Yan Shu[3], Angelos Zavras[4], Ioannis Papoutsis[4], Paolo Rota[3], Begüm Demir[1,2]

BIFOLD[1], TU Berlin[2], University of Trento[3],
National Technical University and National Observatory of Athens[4]

{j.herzog,mathis.juergen.adler,l.hackel,demir}@tu-berlin.de {yan.shu,paolo.rota}@unitn.it
azabras@noa.gr ipapoutsis@mail.ntua.gr

Abstract

Vision-language models (VLMs) have shown strong performance in computer vision (CV), yet their performance on remote sensing (RS) data remains limited due to the lack of large-scale, multi-sensor RS image-text datasets with diverse textual annotations. Existing datasets predominantly include aerial Red-Green-Blue imagery, with short or weakly grounded captions, and provide limited diversity in annotation types. To address this limitation, we introduce `BigEarthNet.txt`*, a large-scale, multi-sensor image-text dataset designed to advance instruction-driven image-text learning in Earth observation across multiple tasks. It contains* 464 044 *co-registered Sentinel-1 synthetic aperture radar and Sentinel-2 multispectral images with* 9.6 M *text annotations, including: i) geographically anchored captions describing land-use/land-cover (LULC) classes, their spatial relations, and environmental context; ii) visual question answering pairs relevant for different tasks; and iii) referring expression detection instructions for bounding box prediction. A comparative statistical analysis shows that* `BigEarthNet.txt` *surpasses existing RS image-text datasets in textual richness and annotation type variety. We further establish a manually-verified benchmark split and extensively evaluate existing VLMs in RS and CV. The results show the limitations of these models on tasks that involve complex LULC classes, whereas fine-tuning using* `BigEarthNet.txt` *results in consistent performance gains across all considered tasks. Our code and data are publicly available at txt.bigearth.net.*

1. Introduction

Unprecedented advances in satellite technology have led to a significant increase in the volume of Earth observation (EO) data archives (*e.g.* the Sentinel satellites of the Copernicus program alone acquire roughly 20 TB of satellite images per day). In this data-rich landscape, natural language provides an intuitive interface for querying and interpreting these vast EO archives, with vision-language models (VLMs) emerging as an effective framework for jointly encoding visual observations and natural language for tasks such as image captioning [6, 12, 33], visual question answering (VQA) [5, 10, 18] and referring expression detection [15, 31, 42].

In general, VLMs are applied to EO data either by using general-purpose computer vision (CV) VLMs directly or by training remote sensing (RS)-specific VLMs. The first approach is limited by the fundamentally different spectral and spatial characteristics between CV and RS images. For example, S2 multispectral (MS) images comprise 13 spectral bands associated with varying spatial resolutions, while those in CV include Red-Green-Blue (RGB) bands in general. The high spectral resolution data enable the discrimination of complex land-use/land-cover (LULC) classes in RS. Although RGB information may be sufficient to distinguish generic classes, *e.g.*,"Water" from "Forests and semi-natural areas", additional spectral information beyond the visible spectrum is needed to differentiate complex LULC classes, *e.g.*, "Mixed forest" vs. "Coniferous forest" [32]. Due to their pretraining phase using RGB-only image-text data, the VLMs in CV have limited capability to effectively query and interpret RS data. Although several RS-specific VLMs trained on RS image-text datasets [2, 15, 25, 42] have recently been developed, most of these models are still trained on RGB data only and thus subject to the above-mentioned limitations. It is worth noting that there are only a few works that include multispectral or multi-sensor (e.g., multispectral, synthetic aperture radar (SAR)) image-text data during pre-training [28, 31]. The use of multi-sensor data is particularly crucial, since different satellite sensors

[*]These authors contributed equally to this work.

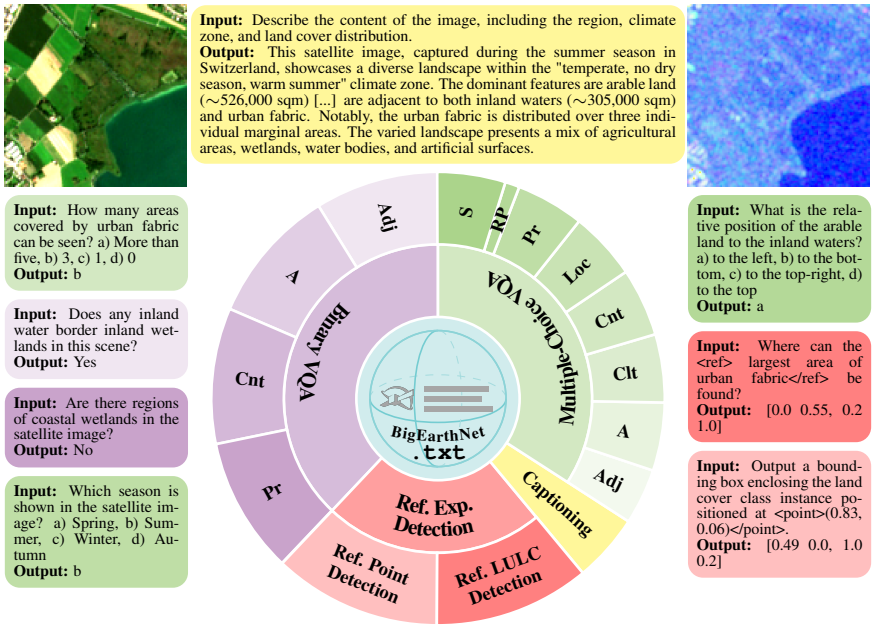


Figure 1. `BigEarthNet.txt` comprises 464 044 co-registered Sentinel-1 (S1) and Sentinel-2 (S2) images with diverse text annotations, resulting in a total of $\sim$ 9.6 million S1-S2-text triplets. The dataset supports 15 tasks (Presence, Area, Counting, Adjacency, Relative Position, Country, Season, and Climate Zone, denoted as Pr, A, Cnt, Adj, RP, Loc, S, and Clt, respectively) across 4 broad categories.

have the capability of measuring different physical properties of LULC classes and their joint use can improve task performance [11, 34, 35]. However, as shown in Tab. 1, the existing RS image-text datasets suffer from: i) limited availability of co-registered multi-sensor data that includes more than three bands, and ii) limited diversity in text annotation types. These limitations directly hinder the ability of VLMs to characterize: 1) the complex spatial/spectral content of RS images; and also 2) complementary information among multi-sensor data.

To overcome the above-mentioned limitations, in this paper we present `BigEarthNet.txt`: a large-scale image-text dataset comprising 464 044 co-registered S1 SAR and S2 multispectral images with 9.6 M text annotations, including captions, VQAs with varying answer formats, and referring expression detection annotations (see Fig. 1). In addition, we introduce a benchmark split that contains 1082 image pairs with 15 029 manually verified text annotations for systematic evaluation of VLMs for EO tasks. Using our benchmark split, we demonstrate that current VLMs in both CV and RS show limited ability to characterize the complex content of multi-sensor RS images. Fine-tuning a state-of-the-art (SOTA) VLM (InternVL3-1B [45]) adapted for multi-sensor input (denoted as InternVL*) on `BigEarthNet.txt` leads to substantial performance gains across all considered tasks. In summary, our contributions are as follows:

1. We show that the existing VLMs from both CV and RS domains exhibit limited capability to characterize the complex spatial/spectral content of multi-sensor images in RS. To this end, we conduct extensive experiments across 15 tasks of 4 categories.
2. We introduce `BigEarthNet.txt`, the first dataset supporting multiple task types, i.e., image captioning, binary VQA, MCQ, and referring expression detection with several different text annotation types for co-registered multi-modal images, with a manually checked subset (benchmark split).
3. We adapt InternVL to accommodate multi-sensor RS inputs and perform domain-specific fine-tuning using `BigEarthNet.txt`. Our results serve as experimental validation of the critical role of large-scale, multi-sensor image-text datasets for accurate interactions with EO data.

2. Related work

VLMs in RS enable natural language interactions with images across multiple tasks, including VQA, change captioning, and referring expression detection [14, 15, 31]. To facilitate the research and development of VLMs, several image-text datasets have been proposed in RS. Early RS image-text datasets, such as UCM-Captions, Sydney-Captions [26], RSICD [20], RSITMD [38], and NWPU-Captions [6] contain high-resolution aerial imagery with human-written single-sentence descriptions with limited intra-class diversity. To overcome this bottleneck, recently

Table 1. Comparison of existing RS image-text datasets. Datasets are compared in terms of: 1) the presence of co-registered multi-sensor imagery exceeding three spectral bands; 2) methods used for text annotation, including the presence of an extensive manual quality check (QC); 3) supported tasks; 4) availability of geolocation data and 5) the number of image-text (IT) samples. (✓) indicates that only parts of the dataset fulfill this property.

Dataset	Multi-Sensor Images			Annotation Method					Supported Tasks			Geolocation data	#IT samples
	Available	>3 bands per image	Co-registered	Manual	Template	Web scraped	LLM-based	Manual QC	Captioning	VQA	Ref. Exp. Detection		
UCM-Captions [26]	✗	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗	✗	10.5k
Sydney-Captions [26]	✗	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗	✗	3k
RSICD [20]	✗	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗	✗	54.6k
RSITMD [38]	✗	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗	✗	23.7k
NWPU-Captions [6]	✓	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗	✗	157k
GAIA [41]	✓	✗	✗	✗	✗	✓	✓	✗	✓	✗	✗	(✓)	205k
Git-10M [17]	✓	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗	✓	10.5M
RSVQAxBEN [19]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	14.8M
RS5M [44]	✓	✗	✗	✗	✗	(✓)	(✓)	✗	✓	✗	✗	(✓)	5.0M
RSTeller [9]	✗	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	2.6M
Landsat30-AU [21]	✗	✗	✗	✗	✗	✗	✓	(✓)	✓	✓	✗	(✓)	216k
ChatEarthNet [39]	✗	✓	✗	✗	✗	✗	✓	(✓)	✓	✗	✗	✗	173k
MS-Clip [23]	✓	✓	✓	✗	✗	✗	✓	✗	✓	✗	✗	✗	985k
`BigEarthNet.txt` (ours)	✓	✓	✓	✗	✓	✗	✓	(✓)	✓	✓	✓	✓	9.6M

proposed datasets shift toward large-scale, automatically generated texts: RS5M [44] introduces the first million-scale corpus with 5 million image-caption pairs; GAIA [41] provides 205 150 pairs, with multi-sentence captions emphasizing environmental dynamics; and Landsat30-AU [21] contains 196 262 image-caption pairs and 17 725 multiple-choice (MC) VQAs. For VQA specifically, RSVQA [18] comprises 77 232 and 1 066 316 question-answer triplets for low- and high-resolution imagery, respectively, while RSVQAxBEN [19] scales to approximately 15 million triplets with S2 BigEarthNet [32] RGB composites and VQAs for CORINE Land Cover (CLC) class presence. Recent studies also extend the spectral range of image-text datasets beyond RGB. Yuan et al. [39] generate captions for nine-band S2 images, yet descriptions rely on broad LULC classes from ESA WorldCover [40] (*e.g.* water, tree, snow). Marimo et al. [23] scale to 975 000 co-registered S1-S2 images from SSL4EO [37] but generate captions using a VLM that receives RGB bands only, preventing texts from relying on information beyond the visible spectrum.

3. BigEarthNet.txt Dataset

To address the scarcity of large-scale multi-sensor image-text datasets in RS, we introduce `BigEarthNet.txt`, which comprises co-registered S1 and S2 images with semantically rich natural-language annotations relevant for 15 tasks across 4 broad categories: i) captioning; ii) binary and iii) multiple-choice VQA; and iv) referring expression detection. The co-registered S1 and S2 images in `BigEarthNet.txt` are taken from BigEarthNet v2.0 [7], which comprises 549 488 pairs of S1 and S2 images acquired over ten European countries. Each pair is accompanied by a pixel-level LULC reference map based on the CLC 2018 product (`V2020_20u1`) [8]. Some images in BigEarthNet v2.0 are either partly covered by seasonal snow, clouds, and cloud shadows, or their respective reference map contains unclassified pixels (with no LULC label). To construct our dataset, we filter out such image pairs, resulting in 464 044 co-registered S1 and S2 images, with an average acquisition time difference of 20 h and a max. of 134 h per image pair. In the following, we describe in detail our annotation generation pipeline, present the caption statistics and lastly introduce our benchmark split.

3.1. Annotation Generation Pipeline

The textual annotations for `BigEarthNet.txt` are generated through a three-stage pipeline (template-based caption generation, large language model (LLM)-based linguistic augmentation, VQA and LULC annotation generation). An overview of the caption generation and linguistic augmentation stage is given in Fig. 2.

Template-based caption generation. We first construct captions by extracting four categories of spatial attributes

directly from the reference maps: the *presence* of LULC classes, the *count* of individual contiguous regions per class (instances), the *size* of each class in total and per instance, and pairwise spatial *adjacency* between classes. Area values are rounded to the nearest $1000\,\mathrm{m}^2$ to reduce label noise. Classes are grouped into three categories according to the proportion of the image they occupy: i) *primary* (> 25 %); ii) *secondary* (5 % - 25 %); or iii) *marginal* (<5 %). Captions are composed from pre-defined templates that contain classes, sizes, and adjacency relations accordingly. Each caption is further grounded in additional context by appending the acquisition season, country, and Köppen-Geiger climate zone [3], derived from high-resolution climate maps.

LLM-based linguistic augmentation. Although template-based captions guarantee factual correctness, they exhibit limited linguistic variety. Therefore, we apply a two-stage augmentation using the quantized Llama-4-Scout-17B model. Initially, a *paraphrasing* step diversifies lexical and syntactic structure while prohibiting the addition of unsupported information. Then, a *self-refinement* step [22] revises the paraphrased output considering the original template to eliminate hallucinated content and restore missing information. To further increase valid lexical variation, the prompt includes the CLC nomenclature, permitting semantically valid substitutions (*e.g.* referring to *Urban fabric* collectively as *Artificial surfaces*) (see supplementary material for used prompts). Manual evaluation of 3209 randomly sampled augmented captions against four binary criteria (linguistic correctness, factual accuracy, completeness, and absence of generation artifacts) yields an average correctness of 93.76 %, with 77.50 % of captions satisfying all four criteria simultaneously.

VQA and referring expression detection generation. We further augment each image pair with question-answer annotations that are relevant for binary yes/no VQA, multiple-choice question (MCQ) VQA, and referring expression detection. Binary questions target the four spatial categories from the captioning stage (presence, count, size, adjacency), with one "yes" and one "no" answer generated per image pair. To prevent "no" answers from being solvable by class-absence detection alone, count and size questions are constructed such that the queried class is present but the stated quantity is incorrect; presence and adjacency "no" questions use semantically similar classes from the CLC hierarchy. MCQs extend the four spatial categories from the captioning stage with questions about relative class position, acquisition country, season, and Köppen-Geiger climate zone. Each set of answers for the MCQs includes one correct answer and three incorrect ones, sampled by analogous principles to the binary VQAs. For referring expression detection, we generate referring LULC detection instructions targeting instances covering between 1 % and 50 % of the image area, and also referring point detection instructions where the center point lies inside the targeted instance. Furthermore, we filter out bounding boxes where the targeted instance covers less than 40 % of the bounding box area. Examples are given in Fig. 1. Each annotation type uses more than 20 linguistic templates to ensure variability without relying on LLM augmentation (see supplementary material C). In total, each image pair is associated with up to 16 VQA pairs, and approximately 80 % of pairs include at least one referring expression detection.

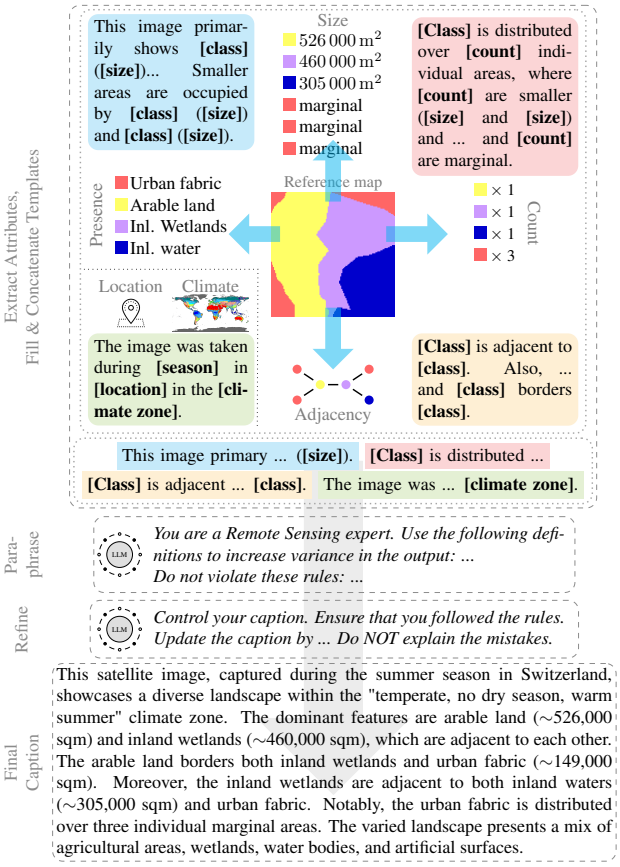


Figure 2. Caption generation process for `BigEarthNet.txt`: i) Attribute extraction from the reference map and template filling based on the attributes and metadata such as location, season and climate zone information, followed by concatenation of the templates; ii) increasing linguistic variance and ensuring correctness via self-refinement.

3.2. Caption Statistics

The caption annotations of `BigEarthNet.txt` contain approximately 50 million words and 2.1 million sentences, averaging 107 words and 4.5 sentences per caption. The vocabulary comprises 12 394 unique terms. Cap-

tion lengths follow a bimodal distribution (Fig. 3a) corresponding to single-class and multi-class scenes: single-class captions are shorter due to the absence of adjacency descriptions, while multi-class captions vary with scene complexity. Sentence counts per caption (Fig. 3b) peak between four and six, reflecting the predominance of complex scenes. As shown in Fig. 3c, the caption annotations of `BigEarthNet.txt` achieve a measure of textual lexical diversity (MTLD) score [24] of 64.69, surpassing the largest existing image-text RS dataset containing images with more than three bands (MS-CLIP [23]) by more than 1.7×. Additionally, caption annotations of `BigEarthNet.txt` encompass ∼25 % more words in half as many samples compared to MS-CLIP. This shows that `BigEarthNet.txt` is not only lexically more diverse, but also richer in the semantic content per caption.

3.3. The BigEarthNet.txt benchmark split

We divide `BigEarthNet.txt` into train, validation and test splits as in BigEarthNet v2.0, resulting in 229 114, 118 095, and 116 835 image pairs and 4 674 281, 2 454 690, and 2 424 991 text annotations, respectively. However, because LLM-augmented captions may contain hallucinated content, we construct a curated benchmark split to enable a reliable model evaluation on the annotations introduced in `BigEarthNet.txt`. The benchmark split contains 15 029 text annotations for 1082 image pairs from the test split whose augmented captions passed manual verification on all four quality dimensions. To expose answer-bias in evaluated models, binary and MCQ annotations are balanced across answer options. Annotations are further balanced across LULC classes to the extent permitted by their natural distribution. Overall, the benchmark split contains 6927 binary and 5550 multiple-choice annotations, 970 captions, and 1582 referring expression detection annotations.

4. Experiments

4.1. Performance Evaluation of VLMs for RS Tasks

To benchmark `BigEarthNet.txt` we have considered a wide range of general-purpose CV and specialized RS VLMs. In detail, we select five SOTA models each from the CV and RS domain. For the CV domain, we select GPT-5.2[1] [30], Qwen3-VL [1], GLM-4.6v-Flash [36], LLaVa-OneVision [16], and InternVL3-1B [45], denoted as GPT, Qwen, GLM, LLaVa, and IVL, respectively. For the RS domain, we select GeoChat [15], LHRS-Bot [25], SkyEyeGPT [42], EarthDial [31], and EarthMind [28], denoted as GC, LHRS, SE, ED_{rgb}/ED_{s2}, and EM_{rgb}/EM_{s1s2}, respectively. The majority of these models accept only RGB inputs; for these, we provide only the RGB bands of S2

[1]OpenAI API, endpoint `/v1/chat/completions`, model "gpt-5.2-2025-12-11", temperature=0, and top_p=0

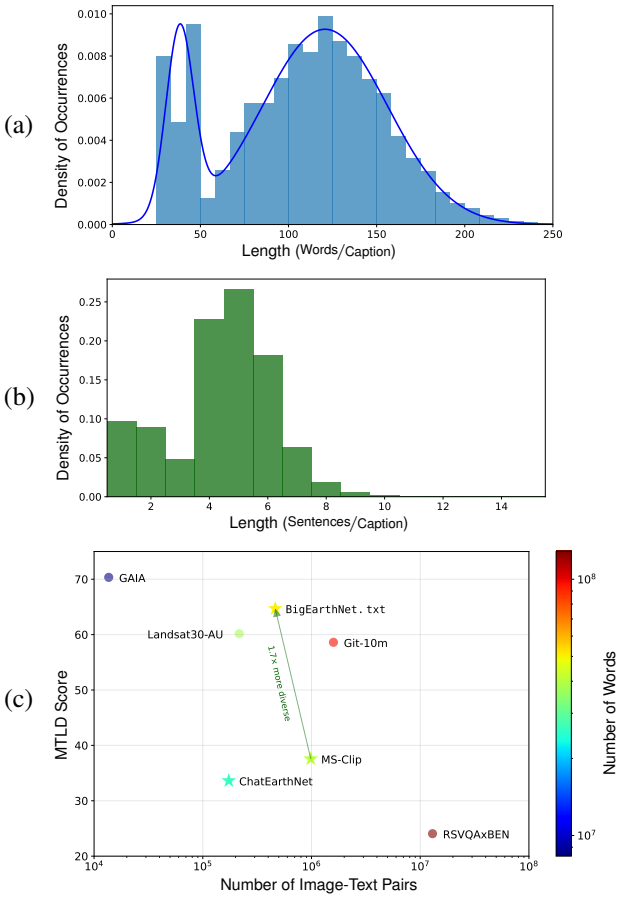


Figure 3. Probability density function (PDF) of (a) caption lengths and (b) number of sentences in a caption in the `BigEarthNet.txt` dataset. (c) Comparison of existing image-text RS datasets in terms of size and semantic richness of the text data. The Natural Language Toolkit (NLTK) word tokenizer is used to divide the captions and questions into tokens. The color of each point indicates the total number of tokens in the dataset. Datasets that encompass images with more than three spectral bands are denoted with a star. The `BigEarthNet.txt` dataset is 1.7× more diverse compared to the largest existing RS dataset encompassing more than three bands.

during inference. As EarthDial and EarthMind also accept multispectral and multi-sensor inputs, we evaluate each under two configurations: i) ED_{rgb} and EM_{rgb} using RGB inputs; and ii) ED_{s2} and EM_{s1s2} using twelve S2 bands and two S1 and twelve S2 bands as input, respectively. We provide the `BigEarthNet.txt` instructions by default, with additional instructions to follow the output format and model specific tokens where necessary. We extract answers even when they do not adhere exactly to the specified format. The final instructions used for each model are provided in supplementary material E. To reduce computational resources during evaluation, we disable thinking

for GLM. Throughout the paper, we report the results on the benchmark split only, whereas additional results on the complete test split of `BigEarthNet.txt` are provided in supplementary material D.

Tab. 2 shows the results obtained for each of our general categories: captioning, binary VQA, MCQ, and referring expression detection, grouped by RS specialized VLMs (top) and general-purpose CV VLMs (bottom). From the table, one can observe that none of the VLMs in general provides good performance, while CV VLMs perform better than RS VLMs in general despite processing only RGB input. This is because CV VLMs exhibit stronger generalizability and instruction-following ability (Tab. 3–6), whereas most RS VLMs are mainly suitable for tasks achievable using only RGB data. The considered CV VLMs may also

Table 2. Results of VLMs in RS and CV on the main categories of the `BigEarthNet.txt` benchmark split. Reported metrics for captioning: BLEU-4, binary VQA: accuracy, MCQ: accuracy, and referring expression detection: mean intersection-over-union (mIoU). All results in percent (%). *: Reported parameter count.

Model	# Parameters	Captioning	Binary VQA	MCQ	Ref. Exp. Detection
GC [15]	7B	0.75	50.82	28.36	4.85
LHRS [25]	7B	0.72	48.23	22.99	4.79
SE [42]	7B	0.00	48.87	27.60	6.08
ED_{rgb} [31]	4B	0.48	58.38	32.94	7.13
ED_{s2} [31]	4B	0.03	44.06	8.43	0.49
EM_{rgb} [28]	4B	**1.66**	57.90	34.25	12.12
EM_{s1s2} [28]	4B	*1.46*	57.79	35.26	16.18
GPT [30]	2T*	0.30	*60.39*	34.93	**31.73**
Qwen [1]	8B	0.70	**61.96**	**37.55**	18.00
GLM [36]	10B	0.79	56.59	34.45	*27.24*
LLaVa [16]	7B	0.96	58.60	*36.27*	20.17
IVL [45]	1B	0.34	54.11	26.76	5.76

have been pretrained using some RS data, which is a growing trend in recent works [29]. Additionally, models accepting multispectral (ED_{s2}) or multi-sensor (EM_{s1s2}) inputs show no consistent benefit (and in some cases decreased performance) compared to their RGB counterparts. This is because their pre-training and fine-tuning data largely consist of RGB images, and the models were therefore not trained to exploit the additional spectral information provided at inference. Even GPT [30], which is the largest model, scores only 60.39% on binary VQA and 34.93% on MCQ due to its inherent RGB data limitation. Tab. 3 shows detailed per-task results for binary VQA. From the table, one can clearly observe that for some tasks, models perform noticeably better (*e.g.*, 69.34 % accuracy for EarthMind on the binary *presence* task). This good result on presence VQA is expected, as determining whether a LULC class appears in an image is closely analogous to scene classification, which is a dominant objective in RS VLM pre-training datasets [13, 15, 25, 28, 31, 42]. Never-

Table 3. Results on the binary VQA tasks for VLMs in RS and CV: Presence, Area, Counting, Adjacency, and Overall, denoted as Pr, A, Cnt, Adj, and OA, in terms of accuracy. All results are in percent (%). IF shows whether an unambiguous answer could be extracted all the time.

Model	Pr	A	Cnt	Adj	OA	IF
GC [15]	51.65	46.86	43.57	55.51	50.82	✓
LHRS [25]	48.83	51.23	49.23	46.10	48.23	✗
SE [42]	52.19	51.61	46.63	46.94	48.87	✓
ED_{rgb} [31]	64.47	53.06	51.99	**60.62**	58.38	✗
ED_{s2} [31]	37.45	40.20	47.17	47.78	44.06	✗
EM_{rgb} [28]	**69.34**	56.20	51.07	55.97	57.90	✓
EM_{s1s2} [28]	*69.07*	54.59	50.15	56.98	57.79	✓
GPT [30]	61.59	**67.38**	*61.94*	55.86	*60.39*	✓
Qwen [1]	64.33	*66.62*	**62.33**	58.45	**61.96**	✓
GLM [36]	60.01	62.40	54.75	53.03	56.59	✓
LLaVa [16]	62.07	58.19	54.36	*58.94*	58.60	✓
IVL [45]	56.24	48.77	51.61	56.60	54.11	✗

theless, none of these models, including the largest, GPT, achieve consistently strong performance across all binary VQA tasks. In Tab. 4, one can observe the performance on the MCQ tasks. These results closely mirror the binary VQA findings: the best performance is again observed on tasks that most closely resemble typical pre-training objectives. Beyond understanding the image content, instruction-following is an additional challenge: most RS models struggle to consistently adhere to the MCQ format, as this task is less common in RS VLM pre-training data and thus less reliably learned. The results on referring LULC detection and referring point detection are presented in Tab. 5 and Tab. 6. Since EarthMind was trained on segmentation rather than bounding box prediction, we derive bounding boxes from their segmentation outputs. In referring LULC detection, models must identify and localize a target instance from a textual description alone, without any spatial prior. In referring point detection, models are additionally provided with a point within the target instance and must predict the enclosing bounding box. As one can see from the table, several models from the CV domain perform substantially better on referring point detection than on referring LULC detection. We think that the provided point allows models with strong general spatial reasoning to constrain the predicted box to the vicinity of the given location, reducing the task to local boundary estimation rather than open-vocabulary instance search. GPT achieves the best performance on both task types, mainly owing to its scale and the breadth of its pre-training data. Finally, Tab. 7 shows the performance obtained in the captioning task in several metrics. We employ n-gram-based (BLEU, ROUGE, METEOR, CIDEr), embedding-based (BERTScore, SBERT-Cosine [27, 43]), and LLM-based (CLAIR [4]) metrics. For CLAIR, an LLM is prompted to output a score between 0

Table 4. Results on the MCQ tasks for VLMs in RS and CV: Presence, Area, Counting, Adjacency, Relative Position, Country, Season, Climate Zone, and Overall, denoted as Pr, A, Cnt, Adj, RP, Loc, S, Clt, and OA, respectively in terms of accuracy. All results are in percent (%). IF shows whether an unambiguous answer could be extracted all the time.

Model	Pr	A	Cnt	Adj	RP	Loc	S	Clt	OA	IF
GC [15]	32.58	26.80	24.72	29.75	25.46	34.62	22.57	30.52	28.23	✗
LHRS [25]	23.84	16.54	21.19	27.23	17.48	28.53	23.71	23.56	22.99	✗
SE [42]	30.89	21.75	24.91	30.51	22.09	29.81	24.71	33.63	27.60	✓
ED_{rgb} [31]	**46.40**	27.41	**36.43**	30.66	29.14	**54.17**	25.57	27.26	32.94	✗
ED_{s2} [31]	21.16	5.51	4.65	8.62	0.00	15.06	6.86	7.26	8.43	✗
EM_{rgb} [28]	41.75	24.81	27.32	42.87	29.91	40.06	23.43	37.04	34.25	✗
EM_{s1s2} [28]	*43.58*	25.57	25.28	**44.70**	31.60	40.06	25.71	36.74	35.26	✗
GPT [30]	38.08	**44.41**	22.68	39.16	27.91	41.35	28.43	34.52	34.93	✓
Qwen [1]	34.41	*37.67*	29.00	42.87	*34.66*	*47.76*	*27.29*	**45.93**	**37.55**	✓
GLM [36]	34.27	32.31	*34.01*	39.44	31.90	38.78	24.71	37.93	34.45	✓
LLaVa [16]	34.56	32.47	32.71	*43.33*	**34.82**	40.38	**27.43**	*39.56*	*36.27*	✓
IVL [45]	28.77	26.03	23.05	30.59	23.01	26.92	24.29	26.96	26.76	✗

Table 5. Results on referring LULC detection for VLMs in RS and CV in terms of mIoU, and accuracy@k (where a prediction is counted as correct if the overlap with the reference is greater than k%). All results are in percent (%). IF shows whether an unambiguous answer could be extracted all the time.

Model	mIoU	Accuracy				IF
		@25	@50	@75	@90	
GC [15]	7.51	11.34	1.51	0.00	0.00	✓
LHRS [25]	7.47	12.59	1.89	0.50	0.00	✗
SE [42]	12.12	20.03	6.55	2.27	0.13	✓
ED_{rgb} [31]	8.99	12.85	2.02	0.00	0.00	✓
ED_{s2} [31]	0.98	0.63	0.00	0.00	0.00	✗
EM_{rgb} [28]	21.90	35.52	*17.88*	**5.67**	**1.39**	✗
EM_{s1s2} [28]	22.65	*38.41*	16.62	*4.53*	1.01	✗
GPT [30]	**25.16**	**40.81**	**19.52**	4.28	*1.26*	✓
Qwen [1]	15.60	25.31	9.95	2.39	0.38	✓
GLM [36]	22.35	37.03	16.75	3.53	**1.39**	✓
LLaVa [16]	*23.04*	*38.41*	15.74	3.90	0.63	✓
IVL [45]	5.01	5.29	0.88	0.00	0.00	✗

Table 6. Results on referring point detection for VLMs in RS and CV in terms of mIoU, and accuracy@k (where a prediction is counted as correct if the overlap with the reference is greater than k%). IF shows whether an unambiguous answer could be extracted all the time.

Model	mIoU	Accuracy				IF
		@25	@50	@75	@90	
GC [15]	2.16	3.43	0.63	0.13	0.00	✗
LHRS [25]	3.09	0.25	0.00	0.00	0.00	✗
SE [42]	0.00	0.00	0.00	0.00	0.00	✗
ED_{rgb} [31]	5.26	3.05	0.13	0.00	0.00	✓
ED_{s2} [31]	0.00	0.00	0.00	0.00	0.00	✗
EM_{rgb} [28]	9.35	14.21	4.57	1.27	0.13	✗
EM_{s1s2} [28]	12.65	17.51	5.08	0.89	0.13	✗
GPT [30]	**38.95**	**64.21**	**35.53**	**8.63**	**2.79**	✓
Qwen [1]	20.42	28.81	8.25	0.89	0.00	✓
GLM [36]	*32.16*	*56.09*	*22.46*	*4.82*	*1.40*	✓
LLaVa [16]	17.27	19.67	3.17	0.51	0.13	✓
IVL [45]	6.52	3.05	0.00	0.00	0.00	✗

and 100 for a candidate caption based on the probability that it describes the same image as the reference caption. We use the distilled reasoning *DeepSeek-R1-Distill-Qwen-32B* with greedy decoding as judge. We observe that all evaluated SOTA VLMs have limited capabilities in describing existing LULC classes and their spatial composition in detail.

In general, these results demonstrate that SOTA VLMs fall short on tasks requiring multi-sensor spectral information and complex LULC understanding. Exceptions arise when a task closely resembles the model's pre-training objectives, *e.g.*, EarthMind's strong performance on binary presence VQA. This suggests that the poor generalization to other `BigEarthNet.txt` tasks is primarily a consequence of insufficient pre-training data, rather than an inherent limitation of the model architectures themselves. Qualitative examples are provided in supplementary material F.

4.2. Results of Fine-tuned InternVL (InternVL*)

Using the training and validation sets of `BigEarthNet.txt`, we fine-tune a SOTA VLM adapted to accommodate multi-sensor input and evaluate it on the benchmark split. To this end, we build on InternVL-3-1B [45] as a backbone and introduce modality-specific branches for S1 and S2 inputs, while we retain the original InternVL components. The model is denoted as InternVL* and its overview is shown in supplementary material G. For each sensor modality, a pretrained vision transformer (ViT) produces patch embeddings, which are aligned to the InternVL LLM embedding space via linear projection layers.

Table 7. Results on the captioning task for VLMs in RS and CV in terms of n-gram-based (BLEU-4, ROUGE, METEOR, CIDEr), sentence-embedding-based (BERTScore, SBert-Cosine) and LLM-based (CLAIR) metrics, denoted as B4, R, M, C, B, SB, and CL respectively. All results are in percent (%).

Model	B4	R	M	C	B	SB	CL
GC [15]	0.75	14.95	14.24	0.65	83.42	43.05	19.91
LHRS [25]	0.72	12.44	16.93	0.01	82.52	*54.61*	28.62
SE [42]	0.00	5.91	2.21	0.00	82.79	38.30	43.86
ED_{rgb} [31]	0.48	13.65	11.36	0.51	83.45	50.49	46.51
ED_{s2} [31]	0.03	8.64	3.51	0.00	83.34	35.00	**59.70**
EM_{rgb} [28]	**1.66**	**16.54**	14.93	**0.96**	**84.34**	50.55	56.73
EM_{s1s2} [28]	*1.46*	*16.43*	14.62	0.76	*84.15*	51.97	56.45
GPT [30]	0.30	13.39	12.69	0.63	83.58	52.52	57.68
Qwen [1]	0.70	11.09	**17.83**	0.01	81.87	52.62	28.38
GLM [36]	0.79	14.94	*17.05*	0.43	83.51	**55.90**	50.91
LLaVa [16]	0.96	14.84	15.41	*0.81*	83.89	52.39	*58.91*
IVL [45]	0.34	11.61	16.71	0.06	81.75	51.22	35.68

The projected S1 and S2 tokens are concatenated with the RGB tokens and the tokenized instruction before being passed to the LLM. To preserve pretrained representations and reduce computational complexity, all ViT backbones are frozen. Only the modality-specific projections and LoRA adapters for the LLM (rank 8, $\alpha = 32$, dropout 0.1) are trained, resulting in 5.8 M trainable parameters out of 1.1 B in total. We initialize the S1 and S2 encoders with BigEarthNet-pretrained ViTs [7], removing their classification heads. We process only the 10 m and 20 m bands of S2, as the 60 m bands are primarily used for cloud screening and atmospheric correction [34]. Training uses batch size 8 with a linear warm-up cosine annealing schedule, increasing the learning rate from 10^{-6} to 10^{-4} over the first 1 % of steps, followed by cosine decay. We fine-tune separately for each task to provide per-task baselines. The model is fine-tuned on the combined training and validation sets for one epoch and evaluated on the `BigEarthNet.txt` benchmark split. Fine-tuning takes approximately two days on four NVIDIA H200 GPUs in total.

Tab. 8 shows the results obtained by InternVL* on the general categories. From the table, one can see that, once a model is fine-tuned on pairs of S1 and S2 data with diverse text annotations, performance on the benchmark split improves substantially across all tasks, even at a small model scale. Further detailed results obtained by InternVL* on all 15 tasks are provided in supplementary material H.

5. Conclusion

In this work, we present `BigEarthNet.txt`, a large-scale multi-sensor RS image-text dataset that encompasses 15 downstream tasks across four distinct categories: image captioning, binary VQA, MCQ, and referring expression detection. Our dataset addresses the critical scarcity of large-scale RS image-text resources spanning diverse textual annotations associated with co-registered multi-sensor data. With this dataset, we also provide a benchmark split, which contains manually verified image-text samples, enabling reliable evaluation of VLM capabilities to model the complex spatial and spectral characteristics of multi-sensor RS data. Through systematic evaluation on the benchmark split, we show that both general-purpose CV and RS-specialized VLMs are subject to limited capabilities. In particular, RS-specific models lead to almost no advantage over their general-purpose counterparts. To demonstrate that these limitations stem from data scarcity rather than architectural constraints, we adapt InternVL-3-1B [45] for multi-sensor data and fine-tune on `BigEarthNet.txt`. The adapted model achieves an improvement of 11.33 % to 32.38 % over SOTA, demonstrating that even a small-scale VLM with simple adaptations can effectively model the complex content of multi-sensor data when an adequate training dataset is considered. In this context, the proposed dataset is very promising for effective and interactive EO, enabling natural language interactions with the complex content of RS images, thus democratizing the use of EO data for non-experts as well.

Table 8. Results on the main tasks of the `BigEarthNet.txt` benchmark split for the fine-tuned adapted InternVL* model as well as best results from VLMs in RS and CV, which are EM_{rgb} [28] and LLaVa [16] for captioning, ED_{rgb} [31] and Qwen [1] for binary VQA, EM_{s1s2} [28] and Qwen [1] for MCQ, and EM_{s1s2} [28] and GPT [30] for referring expression detection. Reported metric for captioning is BLEU-4, while that for binary VQA, and MCQ is accuracy, and that for referring expression detection is mIoU. All results are in percent (%).

Model	Captioning	Binary VQA	MCQ	Ref. Exp. Detection
SOTA RS	1.66	58.38	35.26	16.18
SOTA CV	0.96	61.96	37.55	31.73
InternVL*	**34.04**	**73.29**	**65.84**	**52.41**

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Overview of Supplementary Material

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D. Additional Results on Test Split
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A. Discussions and Broader Impact

In Sec. A.1 we discuss the broader impact of our work and in Sec. A.2 limitations and future work.

A.1. Broader Impact

`BigEarthNet.txt` supports the development and testing of VLMs for RS images with complex content. By providing co-registered multi-sensor data with diverse text annotations describing complex LULC patterns `BigEarthNet.txt` can serve as a valuable fine-tuning resource, as well as part of pre-training a large scale foundation VLM in conjunction with other datasets. The manually-checked `BigEarthNet.txt` benchmark split on the other hand enables the faithful evaluation of pre-trained VLMs. Furthermore, through `BigEarthNet.txt` natural language interaction with RS image archives can be made possible, reducing the barrier for non-experts. This allows the broader public to assess and analyze RS archives and therefore accelerate and democratize several EO applications. Finally, as we publicly released the complete dataset generation pipeline the overall process can easily be adapted and applied to different domains even beyond EO. The generation of almost 500 000 LLM-augmented and self-refined captions as well as the fine-tuning of InternVL* and model inference for SOTA analysis incurs non-trivial computational cost and energy consumption. As the dataset, creation pipeline, and fine-tuned InternVL* checkpoints are publicly released this is a one-time cost.

A.2. Limitations and Future Work

While `BigEarthNet.txt` is a valuable resource for advancements in image-text learning for RS, it has some limitations. `BigEarthNet.txt` is limited in geographical coverage to ten European countries. Consequently, LULC classes that are underrepresented or absent in Europe are not covered. At the moment the co-registered multi-sensor data is limited to two sensors (S1 and S2). In addition, we employ LLMs for rephrasing and CLAIR [4] evaluation for image captioning that could cause hallucinations or nonfactual responses. Our benchmark split counteracts this as it is manually-verified and completely correct. For image captioning, we also report performance under several standard captioning metrics. Finally, while the text annotations of `BigEarthNet.txt` are already diverse in semantic content and type, there exists an inherent limit due to the generation from CLC maps and the 19-class nomenclature. As future work we plan to increase geographical coverage of `BigEarthNet.txt`, consider additional LULC sources, add additional sensing modalities, and increase annotation type variety even further.

B. Prompts for Caption Generation Pipeline

Figure 1 shows the full prompt used for LLM-based paraphrasing of template-based captions. The prompt comprises a general task-description, set of rules to adhere to, in-context examples (shortened for readability), and the actual template-based caption input. Finally we provide the rephrased output from the model.

Prompt for the LLM-based Augmentation - Paraphrase

Task Description

You are a remote-sensing expert. You will be given a static description of a satellite image with well-defined classes. The well-defined classes are enclosed in quotation marks, followed by their area in square meters in brackets. The description also includes information about which classes are bordering or adjacent to one another, ordered by their relative size without repetition. Use varying natural language and sentence structure to rephrase the description into a single, cohesive, well-written remote sensing caption that contains all of the provided information.

Use the following hierarchy definition to further increase the variance in your output:

- "Urban fabric" and "Industrial or commercial units" are part of the more general class "Artificial surfaces".
- "Arable land", "Permanent crops", "Pastures", "Complex cultivation patterns", "Land principally occupied by agriculture with significant areas of natural vegetation", and "Agro-forestry areas" are part of the more general class "Agricultural areas".
- "Broad-leaved forest", "Coniferous forest", "Mixed forest", "Natural grasslands or sparsely vegetated areas", "Moors, heathland or sclerophyllous vegetation", "Transitional woodlands or shrubs", and "Beaches, dunes, or sands" are part of the more general class "Forest and seminatural areas".
- "Coastal wetlands" and "Inland wetlands" are part of the more general class "Wetlands".
- "Inland waters" and "Marine waters" are part of the more general class "Water bodies".

Set of Rules

While writing the caption, do not violate the following rules:

- Use precise language to describe observable features.
- Avoid assumptions or interpretations beyond the given data.
- Do not make assumptions towards the land use and avoid the phrases "land-cover" and "land-use".
- Include the associated area information if you do not change the class names.
- If you group classes, ensure to either remove the associated areas or sum them together and re-evaluate the border information.
- Either choose 5 % or 72'000 sqm when talking about the smaller regions, but do not use both.
- Remove the quotation marks for the predefined classes in your output.
- Do not prefix your output with "Output:".
- If the input mentions the associated climate zone, keep the climate zone definition as is.
- Include the seasonal and country information in the output.

In-context Examples

The following provides a few examples.

Input: The image primarily shows "Arable land" (∼708'000 sqm), followed by "Broad-leaved forest" (∼413'000 sqm). Smaller areas are covered by "Pastures" (∼319'000 sqm) and "Mixed forest" (∼204'000 sqm). "Arable land" borders "Broad-leaved forest" and "Pastures". Moreover, "Broad-leaved forest" is adjacent to "Mixed forest". The "Arable land" is distributed over three individual areas, where one is a larger area (∼525'000 sqm) and two are smaller areas (110'000 sqm and 73'000 sqm). The "Mixed forest" is distributed over two individual smaller areas (116'000 sqm and 88'000 sqm). The image was taken during Summer in Austria. The associated climate zone is "cold, no dry season, warm summer".

Output: This satellite image, captured in Austria during summer, depicts a diverse landscape dominated by agricultural and forested areas. Arable land (708,000 sqm) is most prominent, covering a large area and two smaller areas, and is next to broad-leaved forest (∼413,000 sqm) and pastures (∼319,000 sqm). The expansive broad-leaved forest is adjacent to a mixed forest (∼204,000 sqm), which is spread across two smaller areas, suggesting a varied forest ecosystem. The landscape is characterized by a mix of agricultural, forest and seminatural areas, set within the "cold, no dry season, warm summer" climate zone.

Input: ...

Output: ...

Input: ...

Output ...

Input Template Caption

The image primarily shows "Arable land" (∼526'000 sqm), followed by "Inland wetlands" (∼460'000 sqm). Smaller areas are covered by "Inland waters" (∼305'000 sqm) and "Urban fabric" (∼149'000 sqm). "Arable land" borders "Inland wetlands" and "Urban fabric". Moreover, "Inland wetlands" is adjacent to "Inland waters" and "Urban fabric". The "Urban fabric" is distributed over three individual marginal areas. The image was taken during Summer in Switzerland. The associated climate zone is "temperate, no dry season, warm summer".

This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant features are arable land (∼526,000 sqm) and inland wetlands (∼460,000 sqm), which are adjacent to each other. The arable land also borders urban fabric (∼149,000 sqm), which is scattered across three small, marginal areas. Furthermore, the inland wetlands are next to inland waters (∼305,000 sqm) and urban fabric. The varied landscape presents a mix of agricultural areas, wetlands, water bodies, and artificial surfaces.



Figure 1. Promp for LLM-based caption rephrasing based on template-based version, including general task description, set of rules for the model to follow, three in-context examples (shortened for brevity), actual template-based caption input, and model output. As model we employ LLama-4-Scout-17B.

In Fig. 2 we show the prompt for LLM-based caption refinement. This includes the paraphrase prompt and output from Fig. 1, followed by a refinement instruction asking the model to improve the paraphrased caption based on the initial template-based version and rephrasing instruction.

LLM-based Augmentation - Refinement

Paraphrase Prompt

You are a remote-sensing expert. You will be given a static description of a satellite image with well-defined classes. The well-defined classes are enclosed in quotation marks, followed by their area in square meters in brackets. The description also includes information about which classes are bordering or adjacent to one another, ordered by their relative size without repetition. Use varying natural language and sentence structure to rephrase the description into a single, cohesive, well-written remote sensing caption that contains all of the provided information.
...

This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant features are arable land (∼526,000 sqm) and inland wetlands (∼460,000 sqm), which are adjacent to each other. The arable land also borders urban fabric (∼149,000 sqm), which is scattered across three small, marginal areas. Furthermore, the inland wetlands are next to inland waters (∼305,000 sqm) and urban fabric. The varied landscape presents a mix of agricultural areas, wetlands, water bodies, and artificial surfaces.

Refinement Instruction

Compare your rephrased caption with the initially provided static description. Ensure that your output fulfills the previously defined rules and does not have less information than the original description. Update the caption accordingly and provide the full caption in your output. Do NOT explain what mistake you have found or what you are correcting. Only include the updated caption in your response.

This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant features are arable land (∼526,000 sqm) and inland wetlands (∼460,000 sqm), which are adjacent to each other. The arable land borders both inland wetlands and urban fabric (∼149,000 sqm). Moreover, the inland wetlands are adjacent to both inland waters (∼305,000 sqm) and urban fabric. Notably, the urban fabric is distributed over three individual marginal areas. The varied landscape presents a mix of agricultural areas, wetlands, water bodies, and artificial surfaces.

Figure 2. Prompt used for LLM-based caption self-refinement, including paraphrasing prompt with output from Fig. 1, the refinement instruction and the refined caption output from the model.

C. Instruction Templates

We present examples of the instruction templates used for each task from; i) binary VQA in Tab. 1, ii) multiple-choice VQA in Tab. 2, iii) referring expression detection in Tab. 3, and iv) captioning in Tab. 4. Overall 469 templates are used for `BigEarthNet.txt`.

Table 1. Examples of question templates for each task of the binary VQA category. {...} denotes a placeholder for the actual value depending on the image content.

Task	Templates
Presence	Is/Are any {class} present in the satellite image? Is/Are any {class} shown in the image? Do you see any area(s) of {class} in the satellite image? Are there regions of {class} in the satellite image? Is any portion of the image covered by {class}?
Area	Is the image fully covered by {class}? Do {class} cover {bound} of the image? Are {class} occupying {lower bound} or more of the image? Is the total area of {class} less than {upper bound}? Do {class} cover between {lower bound} and {lower bound} of the image?
Count	Does the image contain exactly one continuous region of {class}? Are {class} present in multiple distinct continuous areas within the image? Does the image show exactly two distinct continuous areas of {class}? Do {class} occupy two or less continuous areas in the image? Would you say that {class} occupy four or more continuous areas in the image?
Adjacency	Does any (instance of) {class} border a/_ {class} in this scene? Is there any (instance of) {class} and a/_ {class} directly connected in the scene? Is any (instance of) {class} directly adjacent to a/_ {class} in this image? Is there direct adjacency between any (instance of) {class} and a/_ {class} in the scene? Does any (instance of) {class} directly border a/_ {class} in this scene?

Table 2. Examples of question templates for each task of the MCQ category. {...} denotes a placeholder for the actual value depending on the image content.

Task	Templates
Presence	Which of these classes is present in the image? a) {class}, b) {class}, c) {class}, d) {class} Which land cover class appears in this satellite image? a) {class}, b) {class}, c) {class}, d) {class} From these options, which class can be observed in the image? a) {class}, b) {class}, c) {class}, d) {class}
Area	How much of the image do {class} cover? a) {area range}, b) {area range}, c) {area range}, d) {area range} What is the total area of {class}? a) {area range}, b) {area range}, c) {area range}, d) {area range} Which range corresponds to the area covered by {class}? a) {area range}, b) {area range}, c) {area range}, d) {area range} How large is the area covered by {class}? a) {area range}, b) {area range}, c) {area range}, d) {area range}
Count	How many continuous areas are covered by {class}? a) {number}, b) {number}, c) {number}, d) {number} Count the separate regions of {class}: a) {number}, b) {number}, c) {number}, d) {number} How many separate areas of {class} can be seen? a) {number}, b) {number}, c) {number}, d) {number}
Adjacency	Identify the adjacent classes: a) {class pair}, b) {class pair}, c) {class pair}, d) {class pair} Which classes are directly connected? a) {class pair}, b) {class pair}, c) {class pair}, d) {class pair} Which pair of classes share a boundary? a) {class pair}, b) {class pair}, c) {class pair}, d) {class pair}
Relative Position	How is/are the {class} relatively positioned to the {class} in the image? a) {rel. pos.}, b) {rel. pos.}, c) {rel. pos.}, d) {rel. pos.} In relation to the {class}, where is/are the {class} located in the image? a) {rel. pos.}, b) {rel. pos.}, c) {rel. pos.}, d) {rel. pos.} What is the relative position of the {class} to the {class}? a) {rel. pos.}, b) {rel. pos.}, c) {rel. pos.}, d) {rel. pos.}
Country	Which country is shown in the satellite image? a) {country}, b) {country}, c) {country}, d) {country} From the options below, select the country shown in the image: a) {country}, b) {country}, c) {country}, d) {country} Choose the country depicted in the image: a) {country}, b) {country}, c) {country}, d) {country}
Season	In which of the following seasons was the satellite image taken? a) {season}, b) {season}, c) {season}, d) {season} Select the season shown in this satellite image: a) {season}, b) {season}, c) {season}, d) {season} In which season was the satellite image captured? a) {season}, b) {season}, c) {season}, d) {season}
Climate Zone	From the following options, pick the one that corresponds to the climate zone shown in the satellite image: a) {climate zone}, b) {climate zone}, c) {climate zone}, d) {climate zone} Identify which of the following climate zones is shown in the satellite image: a) {climate zone}, b) {climate zone}, c) {climate zone}, d) {climate zone} Which climate zone does the satellite image capture? a) {climate zone}, b) {climate zone}, c) {climate zone}, d) {climate zone}

Table 3. Examples of question templates for referring LULC detection and referring point detection. {...} denotes a placeholder for the actual value depending on the image content.

Task	Templates
Referring LULC Detection	Where can the smallest/largest continuous region of {class} be found? Identify the location of the smallest/largest contiguous area of {class}. Where is the smallest/largest continuous area of {class} in the image? Locate a continuous area of {class}. Indicate the position of a continuous region of {class}.
Referring Point Detection	Output a bounding box surrounding the land cover class instance positioned at {(x, y)} in the image. Generate a bounding box enclosing the land cover class instance at {(x, y)} in the satellite image. Create a bounding box around the land cover class instance found at {(x, y)} in the satellite image. Provide a bounding box enclosing the land cover class instance located at {(x, y)} in the image. Output a bounding box surrounding the land cover class instance at {(x, y)} in the satellite image.

Table 4. Examples of question templates for captioning.

Task	Templates
Captioning	Describe this satellite image, noting the region, seasonal context, and distribution of land cover types. Provide a comprehensive description of the scene, including the location, seasonal context and land cover distribution. Describe the observed landscape, specifying the geographic location, season, and land cover composition. Explain what this image shows, including the region, climate zone, and land cover characteristics. Provide an description of the image content, highlighting the geographic location, season, and spatial patterns.

D. Additional Results on Test Set

Additional results for image captioning, binary VQA, MCQ, and referring expression detection on the full `BigEarthNet.txt` test set are detailed in Tab. 5. Due to computational constraints, we evaluate the strongest models from RS (EarthMind) and CV (Qwen3-VL), the backbone model for InternVL* (InternVL-3 (1B)), and the fine-tuned InternVL*. From the table, one can observe that performance on the full test set is mostly similar (within a few %) compared to performance on the `BigEarthNet.txt` benchmark split. The deviation in performance is mainly due to the sampling strategy employed for the benchmark split to allow unbiased and fair evaluation. Accordingly, InternVL*, fine-tuned on the train set shows the largest performance difference. For fair, unbiased, and manually-checked evaluation, we recommend

Table 5. Inference performance of RS and CV VLMs on the main tasks on the `BigEarthNet.txt` test split. Reported metrics for captioning: BLEU-4, binary VQA: accuracy, MCQ: accuracy, and referring expression detection: mIoU. All results are in percent (%).

Model	# Parameters	Captioning	Binary VQA	MCQ	Ref. Exp. Det.
EarthMind$_{s1s2}$ [28]	4B	1.59	55.36	34.73	16.25
Qwen3-VL [1]	8B	0.78	61.27	37.06	18.60
InternVL-3 (1B) [45]	1B	0.52	51.44	27.56	5.94
InternVL*	1B	**31.06**	**80.06**	**74.42**	**51.35**

evaluating on the benchmark split that also drastically reduces computational cost.

E. Full Instructions for Model Evaluation on `BigEarthNet.txt`

Table 6 lists the precise prompts per category used for model evaluation. In general, we only add additional prompts when

Table 6. Per model per category prompt used during inference. {Instruction} denotes the instruction text directly from `BigEarthNet.txt`.

Model	Binary VQA	MCQ	Ref. Exp. Det.	Captioning
GeoChat	{Instruction} Answer with a single word: "yes" or "no".	{Instruction} Answer with exactly one of a, b, c, d.	[refer] {Instruction}	{Instruction}
LHRS-Bot	{Instruction} Answer with a single word: "yes" or "no".	{Instruction} Answer with exactly one of a, b, c, d.	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Coordinates must be in the 224x224 image space. Do not include any other text.	{Instruction}
SkyEyeGPT	[vqa] {Instruction} Answer with a single word: "yes" or "no".	[vqa] {Instruction} Answer with exactly one of a, b, c, d.	[refer] {Instruction}	[caption] {Instruction}
ED_{RGB}	[s2_rgb_10] {Instruction} Answer with a single word: "yes" or "no".	[s2_rgb_10] {Instruction}	[refer] [s2_rgb_10] {Instruction}	[caption] [s2_rgb_10] {Instruction}
ED_{MS}	[s2_ms_10] {Instruction} Answer the question using a single word.	[s2_ms_10] {Instruction} Answer with exactly one of A, B, C, or D.	[refer] [s2_ms_10] {Instruction}	[caption] [s2_ms_10] {Instruction}
EarthMind	{Instruction} Answer with a single word: "yes"or "no".	{Instruction} Answer with exactly one of a, b, c, d.	{Instruction}	{Instruction}
GPT-5.2	{Instruction} Answer with a single word: "yes" or "no".	{Instruction} Answer using either a), b), c), or d).	{Instruction} Answer by only providing the bounding box in the format of: [x_min, y_min, x_max, y_max] where (x_min, y_min) is the top-left corner and (x_max, y_max) is the bottom-right corner and all values are between 0 and 1. The origin of the image is at the top-left corner.	{Instruction} Answer in form of a descriptive, unformatted caption."
Qwen	{Instruction}	{Instruction}	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Values in the ranges of 0 to 1.	{Instruction}
GLM	{Instruction}	{Instruction}	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Values in the ranges of 0 to 1.	{Instruction}
LLaVA	{Instruction}	{Instruction}	{Instruction} Output ONLY the bounding box as [x1, y1, x2, y2]. Values in the ranges of 0 to 1.	{Instruction}
InternVL*	{Instruction}	{Instruction}	{Instruction}	{Instruction}

required. This includes model-specific special tokens (*e.g.* `[refer]` for GeoChat), or instructions to follow the expected output format (*e.g.* `"Answer with a single word:"yes" or "no"."` when the model alone is unable to reliably do so. Additionally, for area VQAs and for the captioning task we add `"The whole image area is 1440000 square meters."` before the instruction, to enable the relation between pixels and areas for each model.

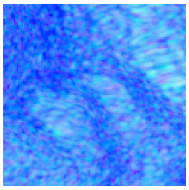
F. Qualitative Examples of Model Answers

Figures 3 to 5 show qualitative per model answers for the binary VQA, MCQ, and referring expression detection tasks. Figure 6 gives an example for the captioning task with RS model answers and Fig. 7 shows the answers of CV models. One can clearly see that given adequate pre-training data a model is capable of answering correctly. Especially in Figs. 6 and 7 it is evident that multi-sensor and diverse pre-training data leads to a more specific description of complex LULC classes (*e.g.* "mixed forest" instead of just "forest"). The considered RS models are GeoChat [15] (denoted as GC), LHRS-Bot [25] (denoted as LHRS), SkyEyeGPT [42] (denoted as SE), EarthDial [31] (denoted as ED_{rgb}/ ED_{s2}), and EarthMind [28] (denoted as EM_{rgb}/ EM_{s1s2}). The considered CV models are GPT-5.2 [30] (denoted as GPT), Qwen3-VL [1] (denoted as Qwen), GLM-4.6v-Flash [36] (denoted as GLM), LLaVa-OneVision [16] (denoted as LLaVa), and InternVL3-1B [45] (denoted as IVL). Further we provide model responses for the fine-tuned InternVL* (denoted as IVL*).

Do regions of any coniferous forest directly adjoin a mixed forest?



S2



S1

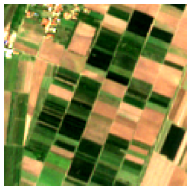
Model	Answer	Correct
GC	yes	✗
LHRS	yes	✗
SE	yes	✗
ED_{rgb}	no	✓
ED_{s2}	yes	✗
EM_{rgb}	yes	✗
EM_{s1s2}	yes	✗
GPT	yes	✗
Qwen	yes	✗
GLM	yes	✗
LLaVa	yes	✗
IVL	no	✓
IVL*	no	✓



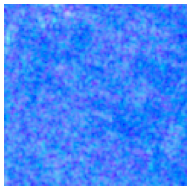
Arable land
Broad-leaved forest
Transitional woodland, shrub

Reference map

Is there only a single continuous region of urban fabrics visible in the image?

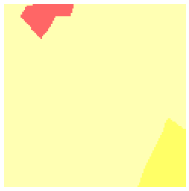


S2



S1

Model	Answer	Correct
GC	no	✗
LHRS	yes	✓
SE	yes	✓
ED_{rgb}	yes	✓
ED_{s2}	no	✗
EM_{rgb}	no	✗
EM_{s1s2}	no	✗
GPT	no	✗
Qwen	no	✗
GLM	yes	✓
LLaVa	no	✗
IVL	no	✗
IVL*	yes	✓



Arable land
Complex cultivation patterns
Urban fabric

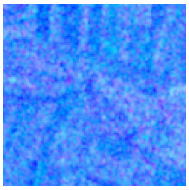
Reference map

Figure 3. Qualitative examples for binary VQA capabilites of considered models. The S2 input is visualized as the true color RGB representation of the S2 image and the S1 input is visualized as (RGB: VV, VH, VV/VH). We provide the reference maps as aid to the reader only. Reference maps are not provided as input to the model.

Identify the country depicted in the satellite image:



S2



S1

a) Austria
b) Switzerland
c) Luxembourg
d) Serbia

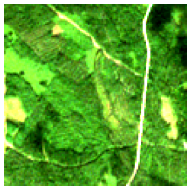
Model	Answer	Correct
GC	a	✗
LHRS	d	✓
SE	c	✗
ED_{rgb}	a	✗
ED_{s2}	a	✗
EM_{rgb}	c	✗
EM_{s1s2}	d	✓
GPT	c	✗
Qwen	a	✗
GLM	c	✗
LLaVa	b	✗
IVL	b	✗
IVL*	d	✓



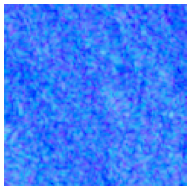
Arable land
Land principally occupied by agriculture, with significant areas of natural vegetation
Transitional woodland, shrub

Reference map

From the following classes, select the one that is present:



S2



S1

a) Broad-leaved forest
b) Coniferous forest
c) Coastal wetlands
d) Inland wetlands

Model	Answer	Correct
GC	a	✗
LHRS	a	✗
SE	a	✗
ED_{rgb}	–	✗
ED_{s2}	d	✗
EM_{rgb}	–	✗
EM_{s1s2}	a	✗
GPT	a	✗
Qwen	a	✗
GLM	a	✗
LLaVa	a	✗
IVL	a	✗
IVL*	b	✓

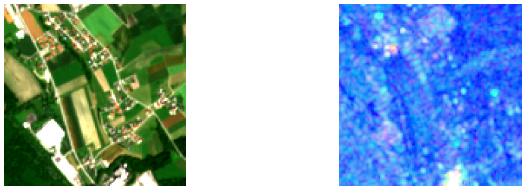


Coniferous forest
Mixed forest

Reference map

Figure 4. Qualitative examples for MCQ capabilities of considered models. The S2 input is visualized as the true color RGB representation of the S2 image and the S1 input is visualized as (RGB: VV, VH, VV/VH). We provide the reference maps as aid to the reader only. Reference maps are not provided as input to the model. "–" indicates that the answer could not be extracted.

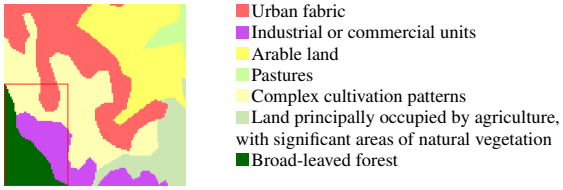
Point out a continuous area of broad-leaved forest in the scene.



S2 S1

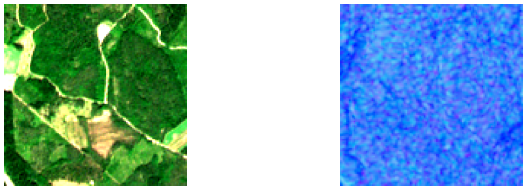
[0.0, 0.44, 0.35, 1.0]

Model	Answer
GC	[0.0, 0.57, 0.24, 0.89]
LHRS	[0.45, 0.45, 0.89, 0.89]
SE	[0.65, 0.19, 0.83, 0.4]
ED_{rgb}	[0.0, 0.79, 0.21, 0.91]
ED_{s2}	–
EM_{rgb}	[0.08, 0.05, 0.52, 0.97]
EM_{s1s2}	[0.74, 0.25, 0.98, 0.63]
GPT	[0.02, 0.29, 0.34, 0.97]
Qwen	[0.0, 0.0, 0.17, 0.25]
GLM	[0.0, 0.0, 1.0, 1.0]
LLaVa	[0.0, 0.67, 0.21, 0.99]
IVL	–
IVL*	[0.0, 0.45, 0.35, 1.0]



Reference map

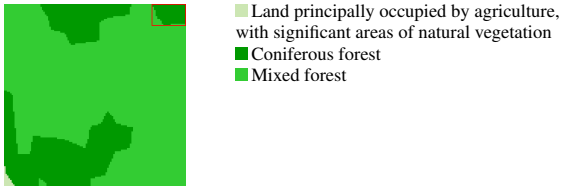
Output a bounding box surrounding the land cover class instance positioned at (0.92, 0.05) in the image.



S2 S1

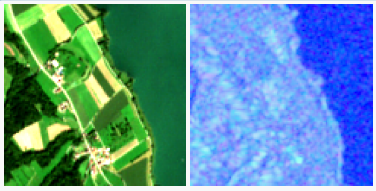
[0.81, 0.0, 1.0, 0.12]

Model	Answer
GC	[0.84, 0.0, 0.96, 0.2]
LHRS	[0.92, 0.05, 0.92, 0.05]
SE	–
ED_{rgb}	[0.06, 0.49, 0.1, 0.55, 0.0]
ED_{s2}	–
EM_{rgb}	[0.0, 0.23, 0.08, 0.92]
EM_{s1s2}	[0.86, 0.22, 0.86, 0.22]
GPT	[0.0, 0.0, 1.0, 1.0]
Qwen	[0.91, 0.04, 1.0, 0.08]
GLM	[0.85, 0.0, 1.0, 0.26]
LLaVa	[0.786, 0.054, 0.998, 0.308]
IVL	[0.93, 0.29, 0.94, 0.61]
IVL*	[0.8, 0.0, 1.0, 0.13]



Reference map

Figure 5. Qualitative examples for referring expression detection capabilities of considered models. The S2 input is visualized as the true color RGB representation and the S1 input is visualized as (RGB: VV, VH, VV/VH). Reference maps are not provided as input to the model. The instance that is to be detected is marked by a red rectangle. "–" indicates that the answer could not be extracted.



S2 S1

Instruction: Explain what this satellite image shows, mentioning the location, season, and landscape features.

Reference: This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant feature is arable land (770'000 sqm), which spans two areas, one large (∼761'000 sqm) and one marginal, covering a small fraction of the total area. The arable land borders a substantial body of inland waters (∼440'000 sqm) and a smaller patch of mixed forest (∼230'000 sqm). The landscape is characterized by a mix of agricultural areas, water bodies, and forest and seminatural areas, highlighting the varied ecosystems present in the region.

EarthMind (RGB) RGB

This satellite image from GoogleEarth shows a rural landscape with a mix of agricultural fields and forested areas. The fields are primarily located in the central and bottom left sections of the image, with a notable concentration of greenery. The forested areas are more prominent in the top left and bottom right sections, with a dense cluster of trees. A water body is visible in the top right corner, adding to the rural character of the scene. The image is taken in a season where the vegetation is in full, indicating a time of growth or harvest. The overall scene is a blend of agricultural land and natural forest, with a water feature adding to the rural setting.

EarthDial (RGB) RGB

The satellite image shows a large area of farmland and a dense forest near the coast. The farmland is located in the center of the image, while the forest is located at the bottom left. The coast is visible at the top right of the image. The image likely represents a coastal region with a mix of agricultural and natural landscapes.

GeoChat RGB

The satellite image shows a diverse landscape with a mix of urban and natural features. The center of the image displays a roundabout, which is likely a busy intersection in a city or town. Surrounding the roundabout are various buildings, indicating a dense urban area. In the background, there are trees and a roundabout, suggesting a well-balanced blend of urban and green spaces. The presence of a roundabout and trees together indicates that the location is likely a well-developed and well-maintained area.

SkyEyeGPT RGB

many farmlands are near a body of water

EarthMind (S1S2) S1,S2

This image shows a landscape with a mix of agricultural fields and forested areas. The fields are primarily located in the central and lower parts of the image, with a notable concentration in the bottom right. The forested areas are more prominent in the upper left and bottom left sections. The image is likely taken in a rural setting, possibly in a region with a combination of agricultural and natural land use. The presence of a water body on the right side suggests proximity to a river or lake, which could be a significant feature for the surrounding land use.

EarthDial (S2) S2

The satellite image shows a land mixed in the left of the image, water in the right of the image.

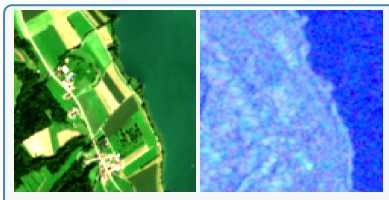
InternVL* RGB,S1,S2

The image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant feature is arable land, spanning approximately 787,000 square meters, which is distributed across two areas, one large (∼786,000 sqm) and one marginal. The arable land borders with inland waters (∼350,000 sqm), mixed forest (∼209,000 sqm), and urban fabric (∼72,000 sqm). The inland waters are adjacent to the mixed forest, which is spread over two areas, one smaller (∼199,000 sqm) and one marginal. The landscape presents a mix of agricultural areas, water bodies, and artificial surfaces, highlighting the varied ecosystems within this region.

LHRS-Bot RGB

Based on the provided image, the location appears to be a rural area with a mix of agricultural lands, forests, and water bodies. The image covers an area of approximately 144,000 square meters.
Location:The image appears to be located in a temperate region, with a mix of flat and hilly terrain. The landscape features rolling hills, valleys, and a large lake in the center of the image.
Season:The image appears to be captured during the summer season, with lush green vegetation and full crops. The trees are full of leaves, and the fields are filled with ripe crops, indicating a warm and sunny weather.
Land Cover Types: 1. Agricultural lands: The image shows a mix of arable lands, pastures, and orchards. The arable lands are cultivated with various crops, including corn, soybeans, and wheat. The pastures are home to livestock, such as cows and sheep. The orchards are filled with fruit trees, such as apple and pear trees. 2. Forests: The image features a mix of deciduous and evergreen forests. The deciduous forests are dominated by trees such as oak, maple, and beech, which shed their leaves in the fall. The evergreen forests are home to coniferous trees such as pine, spruce, and fir. Water bodies: The image features a large lake in the center, surrounded by smaller ponds and streams. The lake is home to a variety of aquatic plants and animals, including fish, birds, and aquatic mammals. 4. Residential areas: The image shows a few small residential areas, likely consisting of single-family homes and farms. 5. Roads and infrastructure: The image features a network of roads, including a main road that runs through the center of the image. Other infrastructure, such as bridges, can be seen in the distance. 6. Recreational areas: The image shows a few small parks and recreational areas, likely used for activities such as hiking, fishing, and boating. Overall, the image depicts a rural landscape with a mix of agricultural lands, forests, water bodies, and residential areas. The season is likely summer, with lush green vegetation and full crops.

Figure 6. A qualitative example of the captioning capabilities of considered RS models. The S2 input is visualized as the true color RGB representation of the S2 image and the S1 input is visualized as (RGB: VV, VH, VV/VH). The input modality for each model is noted in the top-right corner of the respective model answer.



S2 S1

Instruction: Explain what this satellite image shows, mentioning the location, season, and landscape features.

Reference: This satellite image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant feature is arable land (770'000 sqm), which spans two areas, one large (∼761'000 sqm) and one marginal, covering a small fraction of the total area. The arable land borders a substantial body of inland waters (∼440'000 sqm) and a smaller patch of mixed forest (∼230'000 sqm). The landscape is characterized by a mix of agricultural areas, water bodies, and forest and seminatural areas, highlighting the varied ecosystems present in the region.

InternVL3-1B RGB

The satellite image shows a rural landscape with distinct agricultural fields, indicating a countryside setting. Here's an explanation of what the image shows: 1. Location and Season: - The image area spans 144,000 square meters. - Given that it's a remote or aerial shot, it may have been taken in a time of year when the region is often in a different state of cultivation or growth, possibly indicating spring or summer. The vegetation seems lush, typical of a period when crops are actively harvested. 2. Landscape Features: - The landscape includes multiple fields of varying shades of green, with some sections showing more dense vegetation compared to others, possibly indicating different crops or stages of harvest. - There is a noticeable path or road cutting through the fields, running diagonally from the top left to the bottom right. This seems to be a main road, likely connecting different agricultural zones. - The fields are organized in strips and patterns, suggesting structured farming practices. The scattered structures among the fields could be farm buildings, barns, or storage facilities. - The surrounding area has dense greenery, indicating dense forests or possibly a wooded region. This helps in distinguishing the agricultural areas from the forested zones around them. - The water body to the right is likely a large lake, river, or inlet, typical of a riverine or estuarine setting. Its expansive size and darker color contrast with the green fields to the left, suggesting a significant body of water. Overall, the image captures a serene rural landscape with organized agricultural fields, a path for transportation, and natural surroundings suggesting a transitional period between different seasons or stages in agriculture.

GPT-5.2 RGB

A small rural area of patchwork farmland bordered by dense green woodland, likely in a temperate region during late spring or summer, showing rectangular crop fields in varying shades of green and tan, a few scattered farm buildings and houses connected by narrow roads, and sharp field boundaries and hedgerows across the roughly $1.44\,\mathrm{km}^2$ scene.

LLaVA-OneVision RGB

The image shows a rural landscape with a variety of green fields, indicating a lush and fertile area. There is a body of water on the right side, which could be a river or a lake. The fields are divided into different sections, suggesting a mix of agricultural activities such as crop rotation or different types of farming. There are also some structures visible, which could be farmhouses or small buildings, indicating human habitation and activity in the area. The presence of trees and the overall greenery suggest that the season could be spring or summer. The image does not provide enough detail to determine the exact location or the specific types of crops being grown.

GLM-4.6-Flash RGB

The satellite image depicts a rural landscape with a mix of agricultural fields, residential structures, and a body of water (likely a lake or river). The location appears to be a countryside area, possibly in a temperate region given the green vegetation. The season is likely spring or summer, as the fields are lush and green, indicating active vegetation growth. Key landscape features include rectangular agricultural plots (suggesting organized farming), a cluster of buildings (residential or small-scale rural development), and a large water body to the right, which may serve as a natural boundary or resource for the area. The overall scene reflects a blend of human settlement and agricultural land use in a natural setting.

InternVL* RGB,S1,S2

The image, captured during the summer season in Switzerland, showcases a diverse landscape within the "temperate, no dry season, warm summer" climate zone. The dominant feature is arable land, spanning approximately 787,000 square meters, which is distributed across two areas, one large (∼786,000 sqm) and one marginal. The arable land borders with inland waters (∼350,000 sqm), mixed forest (∼209,000 sqm), and urban fabric (∼72,000 sqm). The inland waters are adjacent to the mixed forest, which is spread over two areas, one smaller (∼199,000 sqm) and one marginal. The landscape presents a mix of agricultural areas, water bodies, and artificial surfaces, highlighting the varied ecosystems within this region.

Qwen3-VL RGB

Based on the satellite image and the provided area, here is an explanation of what the image shows: This satellite image displays a rural landscape located in the Czech Republic, specifically in the South Bohemian Region, near the town of Křešice (or possibly the nearby village of Křešice u České Lípy). The image captures a patchwork of agricultural fields adjacent to a large body of water, which is Lake Černé jezírko (Black Lake) or a similar local reservoir. Season: The image was likely taken during late spring or early summer. This is indicated by the vibrant green coloration of the fields, which suggests active growth of crops like cereals or vegetables. The absence of snow or bare soil rules out winter, and the lushness is more characteristic of spring/summer than autumn (when fields would be golden or harvested) or early winter. Landscape Features: - Agricultural Fields: The dominant feature is the grid-like pattern of cultivated land, divided into rectangular plots of varying shades of green and brown. These represent different crops or stages of growth, with some fields appearing darker green (possibly corn or lush grass) and others lighter or brownish (possibly wheat, barley, or fallow land).- Water Body: A large, dark green body of water occupies the right side of the image, indicating a lake or reservoir. Its smooth surface suggests calm conditions.- Roads and Settlements: A network of roads connects the fields and a small cluster of buildings visible near the center-left. These likely represent a rural village or hamlet.- Vegetation: Darker green patches, particularly along the edges of the fields and near the water, indicate forests or wooded areas, which are common in this part of the Czech countryside.- Topography: The terrain appears relatively flat, typical of the Bohemian lowlands, with slight variations in elevation suggested by the contours of the fields and the shoreline. In summary, the image shows a typical Czech rural landscape in late spring or early summer, featuring agricultural fields, a lake, and a small settlement, all within the South Bohemian Region. The 1,440,000 square meter area encompasses this entire scene, highlighting the interplay between agriculture, water, and human habitation.

Figure 7. A qualitative example of the captioning task for considered CV models. The provided input is the same as in Fig. 6. The input modality for each model is noted in the top-right corner of the respective model answer.

G. InternVL* Architecture Overview

The detailed architecture of InternVL* is provided in Fig. 8. We retain the original InternVL components (green), while adding additional paths for S1 and S2 (orange). To leverage InternVL's pre-trained capabilities we provide the RGB bands of S2 to the InternViT. Only the projection layers and LoRA adapters for InternLM are fine-tuned.

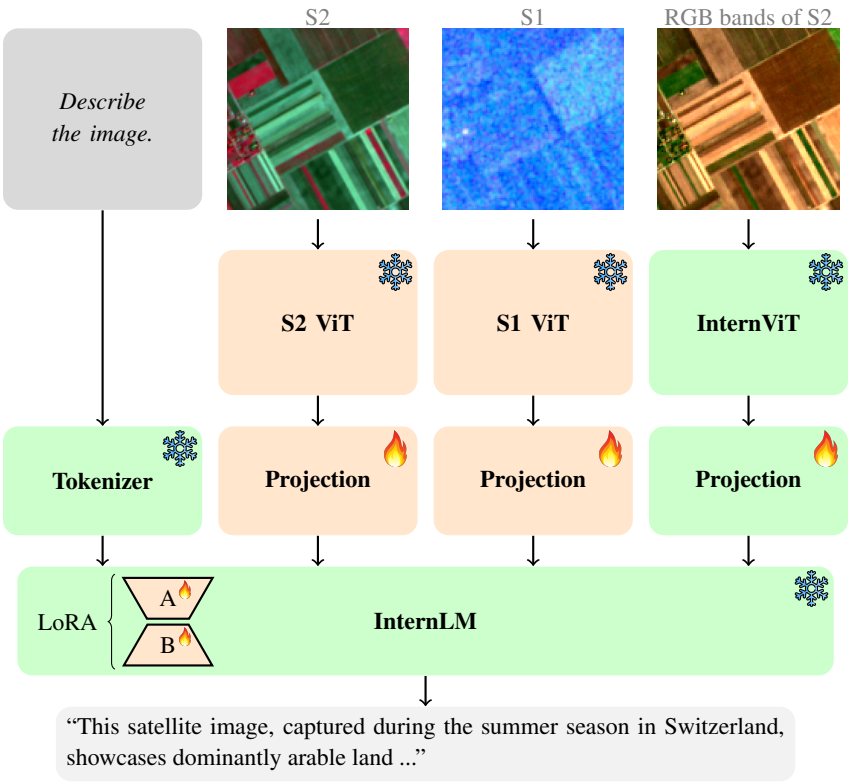


Figure 8. Architectural overview of adapted InternVL to multi-sensor input. InternVL parts are marked in green while additions are orange. The S2 input is visualized as false color composite of an S2 image (RGB: Band 8 – NIR, Band 4 – Red, Band 3 – Green). The S1 input is visualized as (RGB: VV, VH, VV/VH).

H. InternVL* Results for all Tasks

Tables 7 to 11 show results on the benchmark split for the fine-tuned InternVL* on all 15 tasks across the four categories binary VQA (Tab. 7), MCQ (Tab. 8), referring expression detection (Tabs. 9 and 10), and captioning (Tab. 11). We include the top result obtained by an RS model as well as a CV model for each task. One can clearly see that performance is increased drastically when adequate pre-training data is used. Only for the MCQ area task GPT performs slightly better, due to its much larger parameter count. In general, on MCQ GPT performs less favorably.

Table 7. Results on the binary VQA tasks for VLMs in RS and CV: Presence, Area, Counting, Adjacency, and Overall, denoted as Pr, A, Cnt, Adj, and OA, in terms of accuracy from the fine-tuned InternVL* and best results from VLMs in RS and CV. All results are in percent (%).

Model	Pr	A	Cnt	Adj	OA
SOTA RS	69.34	56.20	51.99	60.62	57.90
SOTA CV	64.33	67.38	62.33	58.94	61.96
InternVL*	**69.68**	**82.39**	**78.02**	**68.81**	**73.29**

Table 8. Results on the MCQ tasks for VLMs in RS and CV: Presence, Area, Counting, Adjacency, Relative Position, Country, Season, Climate Zone, and Overall, denoted as Pr, A, Cnt, Adj, RP, Loc, S, Clt, and OA, respectively in terms of accuracy for the fine-tuned InternVL* and best results from VLMs in RS and CV. All results are in percent (%).

Model	Pr	A	Cnt	Adj	RP	Loc	S	Clt	OA
SOTA RS	46.40	27.41	36.43	44.70	31.60	54.17	25.71	37.04	35.26
SOTA CV	38.08	**44.42**	34.01	43.33	34.82	47.76	27.43	45.93	37.55
InternVL*	**64.88**	42.11	**55.20**	**56.90**	**57.98**	**88.78**	**89.14**	**88.44**	**65.84**

Table 9. Results on referring LULC detection for VLMs in RS and CV in terms of mIoU, and accuracy@k (where a prediction is counted as correct if the overlap with the reference is greater than k%). All results are in percent (%).

Model	mIoU	Acc@25	Acc@50	Acc@75	Acc@90
SOTA RS	22.65	38.41	17.88	5.67	1.39
SOTA CV	25.16	40.81	19.52	4.28	1.39
InternVL*	**33.16**	**47.73**	**34.38**	**18.77**	**6.55**

Table 10. Results on referring point detection for VLMs in RS and CV in terms of mIoU, and accuracy@k (where a prediction is counted as correct if the overlap with the reference is greater than k%). All results are in percent (%).

Model	mIoU	Acc@25	Acc@50	Acc@75	Acc@90
SOTA RS	12.65	17.51	5.08	1.27	0.13
SOTA CV	38.95	64.21	35.53	8.63	2.79
InternVL*	**71.81**	**97.21**	**85.66**	**54.19**	**15.74**

Table 11. Results on the captioning task for VLMs in RS and CV in terms of n-gram-based (BLEU-4, ROUGE, METEOR, CIDEr), sentence-embedding-based (BERTScore, SBert-Cosine) and LLM-based (CLAIR) metrics. All results are in percent (%).

Model	BLEU-4	ROUGE	METEOR	CIDEr	BERTScore	SBERT-Cosine	CLAIR
SOTA RS	1.66	16.54	14.93	0.96	84.34	51.97	59.70
SOTA CV	0.96	14.94	17.83	0.81	83.89	55.90	58.91
InternVL*	**34.04**	**50.63**	**50.62**	**78.35**	**91.87**	**83.29**	**70.90**

I. InternVL* Ablations

To evaluate the generalizability of models trained on `BigEarthNet.txt`, we re-train InternVL* with the two largest country splits Serbia and Finland ($\sim$ 47% of train and validation samples) held out and evaluate on them. Table 12 shows the results. When location questions are not evaluated, the accuracy for MCQ reaches 51.58%. This is a reasonable strategy to assess the generalizability, since `BigEarthNet.txt` already covers a large geographical area of 10 countries associated with 19 LULC classes and 10 of 31 global climate zones (see [7] for a detailed discussion of geographic coverage). Additionally, the generalizability of BigEarthNet itself is widely studied, and existing domain adaptation approaches are also applicable for models trained on `BigEarthNet.txt`. To evaluate the role of multispectral over RGB data we ablate InternVL* by fine-tuning only on the RGB bands of the S2 imagery of `BigEarthNet.txt`. Table 13 shows the results. As already shown in [7], jointly using co-registered S1 and S2 images of BigEarthNet improves performance. Similarly, we observe a performance drop for all annotation categories when using only the RGB bands.

Table 12. Results on the main tasks of the `BigEarthNet.txt` benchmark split for the fine-tuned adapted InternVL* model. †: InternVL* fine-tuned on all countries except Serbia and Finland, and evaluated on Serbia and Finland. Reported metric for captioning is BLEU-4, while that for binary VQA, and MCQ is accuracy, and that for referring expression detection is mIoU. All results are given in percent (%).

Model	Captioning	Binary VQA	MCQ		Ref. Exp. Detection
			all	w/o location	
InternVL*†	24.04	62.62	48.31	51.58	40.96
InternVL*	**34.04**	**73.29**	**65.84**	**64.47**	**52.41**

Table 13. Results on the main tasks of the `BigEarthNet.txt` benchmark split for the fine-tuned adapted InternVL* model as well as best results from VLMs in RS and CV, which are EM_{rgb} [28] and LLaVa [16] for captioning, ED_{rgb} [31] and Qwen [1] for binary VQA, EM_{s1s2} [28] and Qwen [1] for MCQ, and EM_{s1s2} [28] and GPT [30] for referring expression detection. InternVL$^*_{rgb}$ fine-tuned and evaluated using only RGB bands. Reported metric for captioning is BLEU-4, while that for binary VQA, and MCQ is accuracy, and that for referring expression detection is mIoU. All results are given in percent (%).

Model	Captioning	Binary VQA	MCQ	Ref. Exp. Detection
SOTA RS	1.66	58.38	35.26	16.18
SOTA CV	0.96	61.96	37.55	31.73
InternVL$^*_{rgb}$	30.88	65.08	55.73	47.78
InternVL*	**34.04**	**73.29**	**65.84**	**52.41**